Lesson Objective

Plot data on line graphs and analyze trends.

Materials: Personal white board

1. Plot data from a table onto a line graph and connect the points.

This table shows how much a puppy named Rosie weighed at different ages.

Age (months)	Weight (pounds)
0	4
3	15
6	25
9	32
12	35

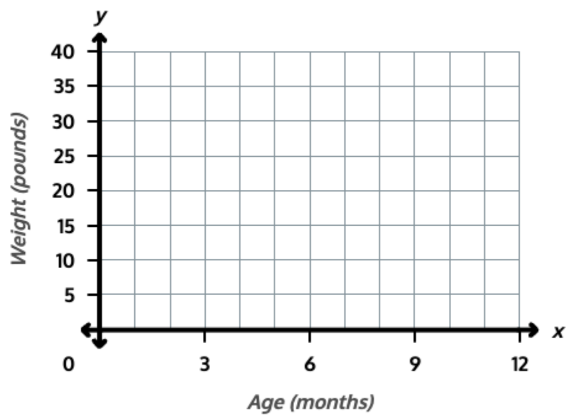
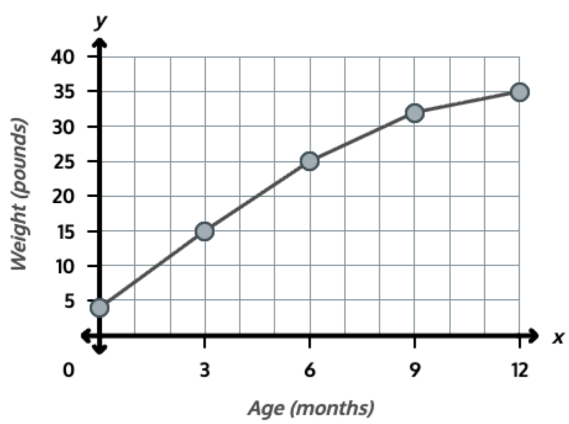
Plot each point from the table on the coordinate plane, then connect the points with line segments to form a line graph.

Show students the table and the coordinate plane. Ask students what the x-axis and y-axis each show and what unit is used on each axis.

Point out that each gridline on the x-axis shows 3 months and each gridline on the y-axis shows 5 pounds.

Model plotting the first point (0, 4) by reading the row in the table, then guide students through plotting (3, 15). Have students plot the remaining points on their own.

Have students connect each pair of consecutive points with a line segment to form the line graph.

 *Answer:* 	
2. Analyze the line graph to find changes between intervals and identify when values stayed constant. Use Rosie's line graph from Problem 1 to answer the questions. a. How much weight did Rosie gain between 0 months and 3 months?	Ask students how to find the weight gained between 0 months and 3 months. Guide them to subtract the y-values: 15 pounds minus 4 pounds equals 11 pounds. Have students confirm by counting up on the y-axis from 4 pounds to 15 pounds.

<table>
<tr>
<td>b. How much weight did Rosie gain between 9 months and 12 months?

c. During which 3-month interval did Rosie gain the most weight?

d. Suppose Rosie weighed 35 pounds at 15 months and 35 pounds at 18 months. What would the line look like between 12 months and 18 months, and what does that tell you about Rosie's weight?

Answer: a) 11 pounds; b) 3 pounds; c) 0 to 3 months; d) The line would be horizontal (flat), showing that Rosie's weight stayed the same.</td>
<td>Have students subtract to find the weight gained between 9 months and 12 months: 35 pounds minus 32 pounds equals 3 pounds.

Ask students to compare the steepness of each line segment to decide which 3-month interval shows the most growth. Point out that the segment from 0 months to 3 months is the steepest, which is why the weight gain there is the greatest.

Ask students what would happen to the line if Rosie's weight stayed at 35 pounds. Guide them to see that a flat, horizontal line shows no change in weight.</td>
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<td>3. Plot a new data set on a line graph and identify the interval with the greatest change by comparing the steepness of line segments.

This table shows the total rainfall during a storm, measured each half hour.

<table>
<tr><th>Time</th><th>Total Rainfall (inches)</th></tr>
<tr><td>3:00 p.m.</td><td>0</td></tr>
<tr><td>3:30 p.m.</td><td>1/2</td></tr>
<tr><td>4:00 p.m.</td><td>1 1/4</td></tr>
<tr><td>4:30 p.m.</td><td>1 1/4</td></tr>
<tr><td>5:00 p.m.</td><td>2</td></tr>
</table></td>
<td>Have students work independently to plot the five points and connect them with line segments.

As students finish, ask them which line segment looks the steepest and what that steepness tells them about the rainfall during that interval.

Have students confirm by subtracting to find the rainfall during the steepest interval: 1 1/4 inches minus 1/2 inch equals 3/4 inch. Confirm this is greater than the change during any other half hour.</td>
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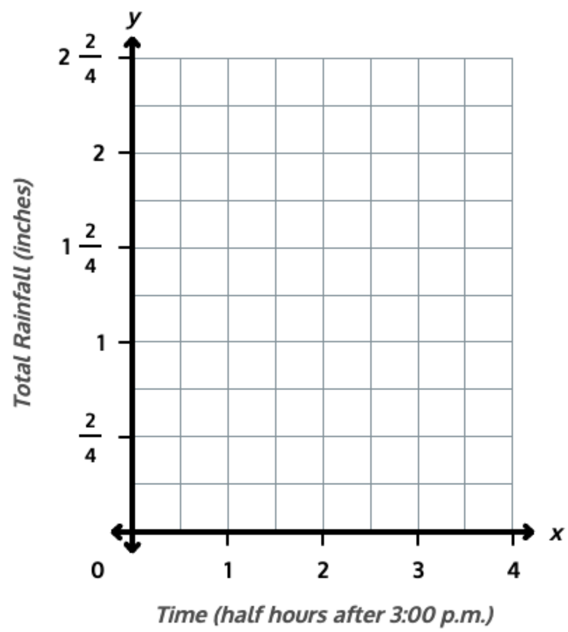
Plot each point on the coordinate plane and connect them with line segments. Then identify the half-hour interval during which the most rain fell.



Answer: 3:30 p.m. to 4:00 p.m. (3/4 inch of rain fell during this half hour, more than any other interval).
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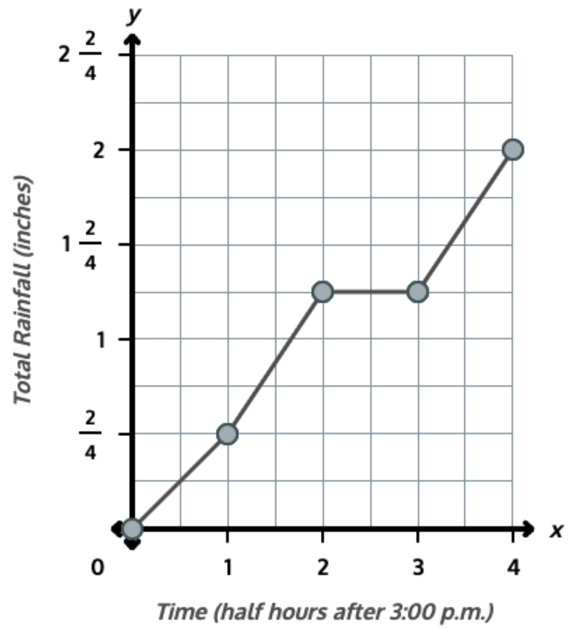

Monitor For	Instructional Moves
• Do students read the scale on each axis correctly, recognizing that one gridline on the x-axis represents 3 months and one gridline on the y-axis represents 5 pounds in Problem 1?	• Before plotting in Problem 1, point to one gridline on the y-axis and say, "This line stands for 5 pounds, not 1 pound." Have students name the value at the next two gridlines to confirm.
• Do students plot each point by matching the x-value and y-value from the table?	• If a student plots a point at the wrong location, prompt them to put a finger on the x-value first, then slide up to the y-value, naming each as they go.
• Do students subtract the y-values at the two endpoints of an interval to find the change, or do they need support recognizing that the difference in y-values is the change in weight or rainfall?	• For Problem 2, if a student is unsure how to find the weight gained, have them count up on the y-axis from one point to the next by 5s.

<table>
<tr><td>• Do students recognize that a horizontal line segment means the value stayed the same?</td><td>• For Problem 3, if a student picks the wrong steepest segment, have them place a finger at the start and end of each segment and compare the vertical rise across the same half-hour width.</td></tr>
<tr><td colspan="2">Reflection Questions

• How did the table help you decide where to plot each point on the line graph? Answer: Each row gave me an x-value and a y-value, so I matched the age to the weight (or time to rainfall) and plotted that ordered pair.

• How did you find the weight Rosie gained between 0 months and 3 months? Answer: I subtracted 4 pounds from 15 pounds to get 11 pounds. I could also count up on the y-axis from 4 to 15.

• What does a horizontal line segment on a line graph tell you about the data? Answer: It means the value stayed the same during that interval — there was no change.

• How did the steepness of a line segment help you find the interval with the most rainfall? Answer: The steepest segment shows the greatest change in the same amount of time, so that interval had the most rain fall.</td></tr>
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